Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/24/2005
PI Name:	Lewis, Richard F. M.D.		
Project Title:	Neural Processing of Ambiguous Gravito-inertial Cues: A Cross Species Comparison		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
PI Email:	richard_lewis@meei.harvard.edu	**Fax:**	FY 617-573-4154
PI Organization Type:	UNIVERSITY	**Phone:**	617-573-3620
Organization Name:	Massachussets Eye and Ear Infirmary/Harvard Medical School		
PI Address 1:	243 Charles Street		
PI Address 2:			
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02114	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	09/01/2001	**End Date:**	08/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Merfeld, Daniel (Massachussets Eye and Ear Infirmary)		
Grant/Contract No.:	NAG 2-1520		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To learn how the nervous system processes ambiguous gravitational and inertial cues such that veridical representations of acceleration and orientation can be calculated by the brain and the appropriate motor and perceptual responses generated.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	This research should help patients with vestibular disorders and normal subjects who experience changes in gravitational force or marked linear accelerations.		

Task Progress:	We have trained two rhesus monkeys to perform a psychophysical task indicating their perception of roll tilt, and have recorded these responses and eye movements during a number of motion profiles that provide ambiguious stimulation to the otolith organs. We have found that perceptual responses in these animals closely correlate with human responses, both in general aspects and in a number of subtle aspects. We have found, for example, that roll tilts were sometimes overestimated (E effects), that perceived tilt decayed when the angle was held (adaptation), and that perceived tilt appeared to depend on a central synthesis of otolith and canal cues. We are currently analyzing eye movement data from these animals in order to compare oculomotor responses with perceptual responses.
Bibliography Type:	Description: (Last Updated: 05/23/2006)
Articles in Peer-reviewed Journals	Lewis, R.F.; Haburcakova, C.; Merfeld D.M. "Tilt psychophysics measured in non-human primates" Annals NY Acad Sci , Mar-2005
Articles in Peer-reviewed Journals	Lewis, R.F. "Frequency-specific mal de debarquement" Neurology , Nov-2004
Books/Book Chapters	Lewis, R.F. "Motion Sickness" in "UptoDate" Ed. Rose, B UptoDate, Jan-2005
Presentation	Lewis, R.F.; Haburcakova, C.; Merfeld, D.M. "Roll tilt perception in rhesus monkeys" Satelllite to the 23rd Barany Society Meeting Jul-2004